HOMEWORK 3

Course

Big Data Analytics

Student

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Report

This report presents the design and implementation of a dimensional data warehouse for the analysis of financial transactions executed during 2024.

The project includes the design of a star schema, the extraction and transformation of data from multiple sources, the resolution of data quality issues, and the execution of analytical queries to generate business insights from transaction data.

1. Introduction

The objective of this project is to design and implement a dimensional data warehouse for the analysis of financial transactions executed during the year 2024.

The project follows a complete Business Intelligence workflow, including data warehouse modeling, data preparation, data quality assessment, data integration, and analytical querying. A star schema was designed to support multidimensional analysis of stock market transactions from different perspectives, including time, geography, company characteristics, and transaction type.

Three datasets were provided for the project:

- A transaction dataset containing BUY and SELL operations executed during 2024.
- A symbols dataset containing information about listed companies, including sector, industry, and country.
- A country dataset containing geographical classifications and regional hierarchies.

The data preparation phase involved identifying and resolving several data quality issues, including missing records, inconsistent country names, and unmatched stock symbols. After cleaning and integrating the datasets, an analytical dataset was created to support business-oriented analyses.

Finally, a set of analytical queries was performed to investigate transaction behavior across different dimensions of the data warehouse and to identify relevant patterns in trading activity during 2024.

2. Data Warehouse Design

2.1 Business Process and Fact Grain

The business process analyzed in this project is the execution of stock market transactions during the year 2024. The available transaction dataset contains BUY and SELL operations performed on publicly traded companies across multiple countries and industries.

The primary objective of the data warehouse is to support multidimensional analysis of trading activity from different perspectives, including time, geography, company characteristics, and transaction type.

A crucial step in dimensional modeling is the definition of the fact grain, which specifies the level of detail represented by each record in the fact table.

For this project, the grain was defined as follows:

One row in the fact table represents one individual transaction recorded in the account statement dataset.

Each transaction contains the following information:

- Transaction identifier.
- Transaction date and time.
- Transaction type (BUY or SELL).
- Stock symbol.
- Number of traded units.

Defining the grain at the transaction level ensures that every operation is preserved and allows analyses to be performed at different aggregation levels through the associated dimensions.

2.2 Fact Table and Dimensions

Based on the identified business process, a star schema was designed consisting of one central fact table and four dimension tables.

The central fact table, **Fact_Transactions**, stores the measurable event of interest, namely the execution of a financial transaction. The main measure contained in the fact table is the number of traded units, while foreign keys connect each transaction to the corresponding dimensions.

Four dimensions were identified:

Dim_Time

The Time dimension supports temporal analysis of transaction activity. It enables users to analyze transactions at different levels of granularity, such as day, month, quarter, and year.

Dim_Symbol

The Symbol dimension contains descriptive information about traded companies, including company name, sector, and industry classification. This dimension supports business-oriented analyses such as identifying the most active sectors or industries.

Dim_Geography

The Geography dimension provides geographical information related to each company, including country, sub-region, and region. This dimension enables geographical comparisons and regional analyses of trading activity.

Dim_TransactionType

The Transaction Type dimension classifies transactions according to their nature, distinguishing between BUY and SELL operations. This dimension allows comparisons between purchasing and selling behavior.

The combination of these dimensions provides a flexible structure capable of supporting a wide range of analytical queries.

Table	**Description**
Fact_Transactions	Stores individual BUY and SELL transactions
Dim_Time	Stores temporal attributes used for time analysis
Dim_Symbol	Stores company, sector, and industry information
Dim_Geography	Stores country, region, and sub-region information

Table	Description
Dim_TransactionType	Stores transaction categories (BUY/SELL)

2.3 Dimension Hierarchies

Hierarchies were defined within the dimensions to support drill-down and roll-up operations typically used in Online Analytical Processing (OLAP).

The Time dimension follows the hierarchy:

Year → Quarter → Month → Day

This hierarchy allows transaction activity to be analyzed at different temporal levels.

The Geography dimension follows the hierarchy:

Region → Sub-Region → Country

This structure enables geographical analyses ranging from broad regional views to country-level details.

The Symbol dimension follows the hierarchy:

Sector → Industry → Company

This hierarchy supports progressive exploration of business information, from broad economic sectors to individual companies.

The presence of these hierarchies increases the analytical capabilities of the data warehouse and facilitates multidimensional exploration of the data.

2.4 Star Schema

The final dimensional model was implemented using a star schema structure.

The fact table occupies the center of the schema and stores transaction events, while the four dimensions provide contextual information required for analysis. The schema was designed to minimize query complexity and improve analytical performance.

The resulting model enables users to analyze financial transactions from multiple perspectives simultaneously, combining temporal, geographical, and business-related dimensions within a single integrated structure.

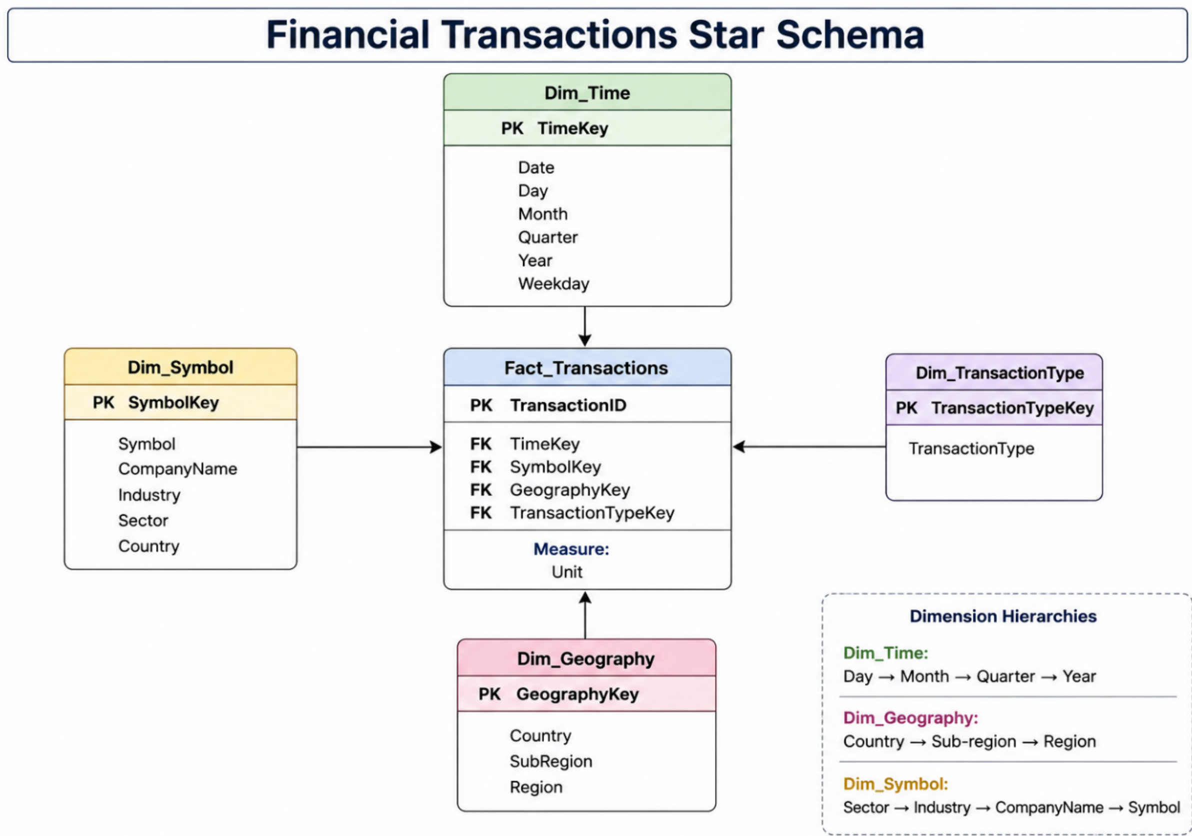

Figure 1. Star Schema designed for the financial transactions data warehouse.

3. Data Preparation

3.1 Data Loading

The first step of the project consisted of loading the three datasets provided for the assignment.

The datasets included:

- A country dataset containing geographical classifications and regional information.
- A symbols dataset containing information about publicly traded companies, including sector, industry, and country.

- A transaction dataset containing BUY and SELL operations executed during 2024.

After loading the datasets into Pandas DataFrames, an initial inspection was performed to verify their structure, dimensions, and attributes.

The datasets contained:

Dataset	**Rows**	**Columns**
Country	249	11
Symbols	3194	5
Transactions	2745	6

This preliminary inspection provided an overview of the available information and served as the basis for subsequent data quality checks.

3.2 Data Quality Assessment

Before integrating the datasets, a data quality assessment was performed to identify potential issues that could affect the analytical results.

The inspection revealed several data quality concerns:

- The transaction dataset contained an unnecessary attribute (Unnamed: 5) that did not provide useful information.
- Missing values were detected in the transaction dataset.
- Some transaction symbols could not be matched to the company reference dataset.
- Country names were not fully consistent across datasets.

Identifying these issues before integration was essential to ensure the reliability and consistency of the final analytical dataset.

3.3 Data Cleaning and Harmonization

The first cleaning operation involved removing the attribute Unnamed: 5, which contained no relevant information for the analysis.

Subsequently, missing values in the transaction dataset were investigated. The analysis showed that the missing values corresponded to 464 completely empty rows located at the end of the file. Since these records did not contain transaction information, they were removed from the dataset.

After removing the empty rows, the transaction dataset contained 2,281 valid transaction records and no remaining missing values.

Additional validation checks were then performed to verify the consistency of relationships between datasets.

A comparison between transaction symbols and the symbols reference dataset identified 18 stock symbols that were not present in the reference table. These transactions were retained because they represented valid trading operations; however, they could not be associated with detailed company information.

A further validation of country names revealed two inconsistencies between the symbols and country datasets:

- Turkey
- Taiwan

A manual inspection showed that these countries were already present in the country reference dataset under different official names:

- Türkiye
- Taiwan, Province of China

To ensure consistency during integration, the country names in the symbols dataset were standardized according to the naming convention adopted in the country dataset.

These cleaning and harmonization procedures ensured that the datasets could be integrated without introducing inconsistencies into the dimensional model.

3.4 Data Integration

After completing the cleaning phase, the datasets were integrated to create a single analytical dataset.

The transaction dataset was first joined with the symbols dataset using the stock symbol as the linking attribute. This operation enriched each transaction with company information such as company name, sector, industry, and country.

The resulting dataset was then joined with the country dataset to incorporate geographical attributes including country, region, and sub-region.

Finally, the transaction date was converted to a datetime format, allowing the derivation of additional temporal attributes required by the Time dimension:

- Year
- Quarter
- Month
- Weekday

The resulting analytical dataset combined transaction, company, and geographical information within a single structure and served as the foundation for all subsequent analytical queries.

4. Analytical Results

The analytical dataset was used to answer a set of business-oriented questions designed to explore transaction behavior from multiple perspectives. The analyses leveraged the dimensions defined in the star schema, including Time, Geography, Symbol, and Transaction Type.

4.1 Top 5 Sectors by Number of SELL Transactions in the United States

The objective of this analysis was to identify the sectors that generated the highest number of SELL transactions among companies headquartered in the United States during 2024.

To answer this question, the analytical dataset was filtered to retain only SELL transactions associated with companies located in the United States. The transactions were then grouped by sector and ranked according to the number of recorded operations.

The results revealed a clear dominance of the Technology sector, which generated 158 SELL transactions. This value is considerably higher than the figures observed for the other sectors included in the ranking. Communication Services ranked second with 58 transactions, followed by Financial Services with 55 transactions, Healthcare with 50 transactions, and Consumer Cyclical with 48 transactions.

The substantial gap between Technology and the remaining sectors suggests that trading activity was heavily concentrated in technology-related companies. This outcome highlights the importance of the technology industry within the analyzed portfolio and reflects the strong presence of major technology firms in the U.S. stock market.

Overall, the results indicate that the Technology sector played a central role in SELL activity during 2024 and represented the primary contributor to transaction volume among U.S.-based companies.

4.2 Top 5 Industries by Number of BUY Transactions in Q4 2024

The second analysis focused on identifying the industries that attracted the highest number of BUY transactions during the fourth quarter of 2024.

The analytical dataset was filtered to include only BUY transactions executed during Q4 2024. Transactions were then grouped by industry and ranked according to their frequency.

The results showed that the Internet Content & Information industry recorded the highest number of BUY transactions, with a total of 15 operations. The Semiconductors industry followed with 13 transactions, while Software - Infrastructure ranked third with 10 transactions. Internet Retail and Diagnostics & Research completed the top five industries with 8 and 7 transactions, respectively.

A common characteristic among the leading industries is their strong connection to technology, digital services, and innovation-driven business models. This pattern suggests that investor interest during the final quarter of 2024 was primarily concentrated in sectors associated with technological development and digital transformation.

The results therefore indicate that technology-oriented industries represented the most attractive investment targets during Q4 2024, confirming the central role of innovation-related companies within the analyzed trading activity.

4.3 Top 10 Industry–Region Pairs by Number of Transactions

The purpose of this analysis was to identify the combinations of industries and geographical regions that generated the highest transaction activity during 2024.

Unlike the previous analyses, which focused on a single dimension at a time, this query combined information from both the Symbol and Geography dimensions. By analyzing industry–region pairs, it was possible to obtain a more comprehensive view of how transaction activity was distributed across different economic sectors and geographical areas.

The results showed that the pair Semiconductors – Americas ranked first with 134 transactions. It was closely followed by Software - Infrastructure – Americas with 132 transactions and Internet Content & Information – Americas with 119 transactions.

Several additional combinations from the Americas also appeared among the top positions, including Telecom Services, Software - Application, Biotechnology, and Credit Services. This pattern highlights the dominant role of the Americas region in overall transaction activity.

The ranking also included important European industry–region pairs. In particular, Semiconductors – Europe generated 100 transactions, while Biotechnology – Europe recorded 66 transactions. Furthermore, Auto Manufacturers – Asia appeared among the top ten combinations with 60 transactions, indicating a significant contribution from Asian companies operating in the automotive sector.

A notable observation is the strong presence of technology-related industries throughout the ranking. Semiconductors, Software, Internet Services, and

Telecommunications collectively account for a substantial share of total transactions. This suggests that investor activity was concentrated in sectors characterized by innovation, digital transformation, and technological development.

Overall, the analysis demonstrates that the highest levels of trading activity were generated by technology-oriented industries located in the Americas, while Europe and Asia contributed through specific high-performing sectors such as Biotechnology and Automotive Manufacturing. The results confirm the usefulness of the multidimensional model in identifying patterns that would not be visible when analyzing industries or regions separately.

4.4 Top 10 Countries by Number of SELL Transactions

The fourth analysis focused on the geographical distribution of SELL transactions in order to identify the countries associated with the highest trading activity during 2024.

To perform this analysis, the analytical dataset was filtered to include only SELL transactions. The transactions were then grouped by country and ranked according to their frequency.

The results showed a strong concentration of SELL transactions in the United States of America, which recorded 389 transactions and ranked first by a considerable margin. The United Kingdom of Great Britain and Northern Ireland occupied the second position with 130 transactions, while China ranked third with 112 transactions.

Other countries included in the top ten were Brazil, the Netherlands, Switzerland, Ireland, Luxembourg, Canada, and Germany. Although these countries contributed significantly to transaction activity, their values remained substantially lower than those observed for the United States.

The large gap between the United States and the remaining countries suggests that a substantial portion of the analyzed portfolio was concentrated in U.S.-based companies. This result is not surprising given the global importance of U.S. financial markets and the large number of technology and growth-oriented companies headquartered in the country.

At the same time, the presence of countries from Europe, Asia, and Latin America demonstrates a certain degree of geographical diversification within the portfolio. The results therefore indicate that while transaction activity was strongly centered on the United States, investors also maintained exposure to companies operating in multiple international markets.

Overall, this analysis highlights the dominant role of the United States in SELL activity during 2024 while confirming the international nature of the analyzed transaction dataset.

4.5 Top Regions by Total Units Bought

The final analysis examined the geographical distribution of purchased trading volume by identifying the regions that accumulated the highest number of bought units during 2024.

Unlike the previous analyses, which focused on the number of transactions, this query considered the total quantity of traded units. Consequently, it provides a different perspective on investor behavior by measuring trading volume rather than transaction frequency.

The results revealed that the Americas generated the highest purchased volume, with a total of 37,026 units. This value significantly exceeded the volume recorded by the other regions and represented the largest share of BUY activity during the year.

Europe ranked second with 22,528 units, while Asia accounted for 9,198 units. Although Europe contributed substantially to trading volume, the Americas maintained a clear leadership position. Asia, while less prominent in terms of purchased units, still represented an important component of the overall portfolio.

These findings are consistent with the results of previous analyses, which repeatedly highlighted the importance of companies located in the Americas. The strong performance of this region can be attributed to the large number of transactions involving companies operating in highly active sectors such as Technology, Software, Internet Services, and Telecommunications.

From a business intelligence perspective, this analysis demonstrates that the Americas not only generated the highest transaction counts but also concentrated the largest trading volumes. Therefore, the region played a central role in shaping overall investment activity throughout 2024.

In conclusion, the geographical analysis of purchased units confirms the dominant position of the Americas within the analyzed dataset, while Europe and Asia provided additional diversification and contributed to the international scope of the investment portfolio.

5. Conclusions

This project demonstrated the complete design and implementation of a dimensional data warehouse for financial transaction analysis.

A star schema was developed to model transaction activity, integrating temporal, geographical, and company-related dimensions. Several data quality issues were identified and resolved during the ETL process, including missing records, inconsistent country names, and unmatched stock symbols.

The analytical queries highlighted significant patterns in trading activity during

2024. The results showed a strong concentration of transactions in technology-related sectors and industries, particularly within the Americas region. Furthermore, the United States emerged as the dominant country in terms of SELL transactions, while purchased trading volume was primarily concentrated in companies located in the Americas.

Overall, the dimensional model proved effective in supporting multidimensional analysis and generating business insights from financial transaction data.